

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Reverse Engineering Task

For each task, study the described big data solution, infer how it was built, identify the underlying components, and explain what design decisions were likely made.

1. A recommendation engine processes user clicks, search queries, and purchases from multiple devices.
Reverse engineer the probable data types, ingestion flow, and analytics stages.
2. A ride-sharing company displays live traffic heatmaps and demand forecasts. Reverse engineer the likely distributed storage and processing backbone behind the system.
3. A fraud detection dashboard flags suspicious card transactions in near real time. Reverse engineer the probable big data security, streaming, and alerting mechanisms used.
4. A social media platform stores videos, text comments, hashtags, and user engagement metrics.
Reverse engineer how structured and unstructured data are probably managed together.
5. A manufacturing IoT platform collects machine logs and maintenance events from many plants.
Reverse engineer the likely integration tools, storage strategy, and analytics path.

Reverse Engineering Task Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Inference Accuracy	25%	Infers system components and flow with high accuracy	Mostly accurate inference	Basic inference	Weak inference
Understanding of Architecture	25%	Shows clear understanding of underlying big data architecture	Good understanding	Partial understanding	Poor understanding
Use of Technical Evidence	20%	Uses strong reasoning and technical evidence	Reasonable evidence	Limited evidence	No meaningful evidence
Depth of Analysis	15%	Analysis is deep and insightful	Reasonably deep	Basic depth	Superficial analysis
Clarity	15%	Clear and organized explanation	Generally clear	Somewhat unclear	Poorly presented

ANSWER:

1. Recommendation Engine

A recommendation engine processes user clicks, search queries, and purchases from multiple devices. The probable architecture can be reverse engineered as follows:

Probable Data Types

- Structured: User IDs, product IDs, purchase details, timestamps.
- Semi-structured: Clickstream logs, JSON events, search queries.
- Unstructured: Product reviews, images, and text.

Probable Data Flow

Mobile/Web Devices



Data Collection & Ingestion



Message Queue / Streaming System



Distributed Storage



Data Cleaning & Processing



User Behavior Analytics



Recommendation Model



Personalized Recommendations

Likely Design Decisions

- Streaming ingestion is used because clicks and searches arrive continuously.
- Distributed storage is used because data comes from millions of users.
- User IDs are used to combine activity from multiple devices.
- Historical purchases and recent clicks are analyzed together.
- Machine-learning algorithms generate personalized product recommendations.

Inference: The system is likely designed as a scalable pipeline combining real-time user activity with historical data.

2. Ride-Sharing Live Traffic and Demand Forecasting

The ride-sharing company needs to display live traffic heatmaps and predict future ride demand.

Probable Architecture

GPS + Driver Apps + Passenger Apps + Traffic Sensors

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Real-Time Data Ingestion

↓

Distributed Storage

↓

Stream Processing

↓

Real-Time Traffic Analytics

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Heatmap Generation

At the same time:

Historical Trip Data

↓

Distributed Storage

↓

Batch Processing & Machine Learning

↓

Demand Forecasting

↓

Driver/Passenger Applications

Distributed Storage

A distributed file system or cloud data lake can store:

- GPS locations
- Trip records
- Traffic information
- Historical demand data

Data is distributed across multiple machines, allowing the system to handle large volumes and failures.

Processing Backbone

- Stream processing: Handles current GPS and traffic data.
- Batch processing: Analyzes historical trips.
- Parallel processing: Processes data across multiple machines.
- Caching: Stores frequently accessed results for faster heatmap display.

Inference: The system probably uses a combination of distributed storage, real-time stream processing, and batch analytics to provide both live traffic information and demand forecasts.

3. Real-Time Fraud Detection Dashboard

A fraud detection system monitors card transactions and identifies suspicious activity in near real time.

Probable Data Flow

Card Transactions



Secure Data Ingestion



Streaming Platform



Real-Time Processing



Fraud Detection Rules/Model



Risk Score



Alert System



Fraud Dashboard

Streaming Mechanism

A streaming platform can continuously receive transaction events. A stream-processing engine analyzes each transaction immediately.

It may check:

- Unusual transaction amount.
- Multiple transactions within a short period.
- Unusual location.
- Sudden change in spending behavior.

- Repeated failed transactions.

Security Mechanisms

- Encryption during data transmission and storage.
- Strong authentication.
- Role-based access control.
- Data masking for card information.
- Audit logs.
- Secure APIs.
- Monitoring of suspicious access.

Alerting

When a transaction exceeds a predefined risk threshold:

Transaction → Risk Analysis → High Risk → Alert → Security Team/Customer

Inference: The system is likely designed with secure streaming ingestion, real-time analytics, automated risk scoring, and immediate alert generation.

4. Social Media Data Management

A social media platform handles videos, comments, hashtags, and user engagement metrics, which contain different types of data.

Probable Data Types

Data	Type
User ID, likes, shares	Structured
Hashtags and JSON events	Semi-structured
Comments	Unstructured/Text
Videos and images	Unstructured

Probable Architecture

Users



Data Collection



Data Integration & Ingestion



Data Lake / Distributed Storage



Structured DB | Object Storage | Distributed Storage



Processing & Analytics



Recommendations / Trends / Reports

Management Approach

- Structured engagement data can be stored in databases.

- Videos and images can be stored in scalable object storage.
- Comments and hashtags can be processed using text analytics.
- Metadata such as video ID, uploader ID, and timestamp can be stored separately.
- Distributed processing can analyze large-scale user activity.

Likely Design Decisions

The platform separates large media files from their metadata and engagement information while maintaining links between them using IDs.

Inference: A combination of relational/NoSQL databases, object storage, distributed processing, and data lakes is likely used to manage different data formats efficiently.

5. Manufacturing IoT Platform

The manufacturing company collects machine logs and maintenance events from many plants. Since machines continuously generate data, the system requires scalable ingestion and storage.

Probable Data Flow

Machines & IoT Sensors

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IoT Gateway

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Integration / Message Streaming Tools

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Distributed Storage / Data Lake

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Data Cleaning & Processing

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Analytics & Machine Learning

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Maintenance Prediction

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Maintenance Alerts

Integration Tools

Integration tools are likely used to:

- Collect data from machines at different plants.
- Connect different machine protocols and systems.
- Transform data into a common format.
- Combine machine logs with maintenance records.
- Transfer data to centralized storage.

Storage Strategy

A data lake or distributed file system can store:

- Sensor readings.
- Machine logs.
- Maintenance records.
- Error messages.
- Historical production data.

Frequently accessed operational data may also be stored in a database for faster querying.

Analytics Path

Historical machine data → Data cleaning → Pattern analysis → Predictive analytics → Failure prediction → Maintenance scheduling.

For example, if temperature and vibration readings continuously increase before previous machine failures, analytics can identify this pattern and generate an early maintenance alert.

Inference: The platform likely combines IoT gateways, integration tools, distributed storage, stream/batch processing, and predictive analytics to enable proactive machine maintenance.